

VAIDSYS TECHNOLOGIES

Embedded Systems & IoT Internship Program

May - June 2026

IoT-Based Home Automation System

NodeMCU ESP8266 · DHT22 · OLED · Relay · RGB LED · Blynk IoT

Intern Name	Utkarsh Saumya
Internship Domain	Embedded Systems + IoT
Project Title	IoT-Based Home Automation System
Microcontroller	NodeMCU ESP8266 (AMICA CP2102)
Sensor	DHT22 — Temperature & Humidity (±0.5°C)
Display	1.3" SH1106 OLED (I2C, 128×64 px)
Actuator	Single Channel Relay Module (5V, Active LOW)
Indicator	RGB LED Module (Common Cathode)
IoT Platform	Blynk IoT (MQTT over WiFi)
Organization	Vaidsys Technologies
Submission Date	June 21, 2026

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1. Executive Summary

This project presents a fully functional IoT-Based Home Automation System built on the NodeMCU ESP8266 platform. The system enables remote monitoring and control of home appliances via the Blynk IoT mobile application, with intelligent temperature-based auto-control powered by the DHT22 sensor.

The system was physically assembled and tested on a dedicated demonstration board. All components — OLED display, DHT22 sensor, relay module, and RGB LED indicator — were individually verified before integration. The project demonstrates a complete IoT development lifecycle: hardware design, embedded firmware development, cloud integration, and mobile app configuration.

Key achievements of this project:

- Live temperature and humidity monitoring with $\pm 0.5^{\circ}\text{C}$ accuracy using DHT22
- Remote appliance ON/OFF control via Blynk IoT mobile and web dashboard
- Intelligent auto-mode: appliance activates automatically when temperature exceeds threshold
- Real-time OLED display showing sensor data, relay state, and operating mode
- Color-coded RGB LED status indicator for instant visual feedback
- Physical demonstration board with labeled components for professional presentation

2. Introduction

2.1 Background

The Internet of Things (IoT) refers to the network of interconnected physical devices that communicate and exchange data over the internet. With over 15 billion connected devices globally as of 2024, IoT has become a cornerstone of modern technology infrastructure. Home automation is among the most impactful consumer applications of IoT, enabling residents to remotely monitor and control household systems from anywhere in the world.

The NodeMCU ESP8266 has emerged as the de facto platform for beginner and intermediate IoT projects due to its built-in WiFi capability, low cost (under ₹250), compatibility with the Arduino development ecosystem, and a large community of developers. This project leverages the ESP8266 to build a production-quality home automation prototype.

2.2 Project Context

This project was developed as part of the Vaidsys Technologies Embedded Systems & IoT Internship Program. The objective was to design, build, and document a complete IoT system from scratch — covering hardware assembly, embedded firmware, cloud connectivity, and user interface design.

The system was physically built on a demonstration board using real hardware components, providing hands-on experience with the complete IoT development stack rather than a simulation-only approach.

3. Problem Statement

Conventional residential and commercial buildings suffer from the following limitations:

- Appliances must be manually operated, even during off-hours or when residents are away.
- There is no real-time visibility into environmental conditions such as temperature and humidity.
- Energy is wasted when appliances (fans, ACs, heaters) are left running unnecessarily.
- No automated response mechanism exists for environmental triggers such as rising temperature.
- Remote monitoring and control from mobile devices is not possible with conventional switches.

3.1 Proposed Solution

This project addresses all the above problems by implementing a WiFi-connected smart home system that:

- Provides real-time environmental monitoring via the DHT22 sensor
- Enables remote appliance control through the Blynk IoT mobile application
- Automatically activates appliances when temperature exceeds a configurable threshold
- Displays live system status on a 1.3-inch OLED display
- Provides immediate visual feedback through a color-coded RGB LED indicator

4. Objectives

1. Design and implement a complete IoT home automation system using NodeMCU ESP8266.
2. Integrate DHT22 sensor for accurate real-time temperature ($\pm 0.5^{\circ}\text{C}$) and humidity monitoring.
3. Enable bidirectional remote control via Blynk IoT mobile and web dashboard over WiFi.
4. Implement intelligent auto-control: automatic appliance activation based on temperature threshold.
5. Display live system status on a 1.3-inch SH1106 OLED display, updating every 2 seconds.
6. Provide color-coded visual feedback via RGB LED module for instant status recognition.
7. Physically assemble and label all components on a professional demonstration board.
8. Document the complete project with circuit diagrams, code explanation, and test results.

5. Tools & Technologies

5.1 Hardware Components

#	Component	Technical Specification	Role in Project
1	NodeMCU ESP8266	AMICA CP2102, Tensilica L106 @ 80MHz, 3.3V, 802.11 b/g/n WiFi, 4MB Flash	Main microcontroller + WiFi
2	DHT22 Sensor	Temp: -40 to +80°C (±0.5°C), RH: 0-100% (±2-5%), Single-wire protocol	Environmental sensing
3	1.3" OLED Display	SH1106 driver, I2C (0x3C), 128×64 px, 3.3V, 0.96W	Real-time status display
4	Relay Module	5V single channel, Active LOW, 10A/250VAC, optocoupler isolated	Appliance switching
5	RGB LED Module	Common cathode, built-in current limiting resistors, 3-channel	Visual status indicator
6	Breadboard	840-point solderless, mounted on demo board	Circuit prototyping
7	Jumper Wires	M2M, M2F sets — color-coded	Component interconnection

5.2 Software & Development Tools

Tool / Platform	Version	Purpose
Arduino IDE	2.3.10	Firmware development, compilation, upload
Blynk IoT Platform	1.3.2	Cloud dashboard, mobile app, MQTT broker
U8g2 Library (oliver)	2.36.19	SH1106 OLED graphics rendering
DHT Library (Adafruit)	1.4.6	DHT22 sensor data acquisition
Adafruit Unified Sensor	1.1.14	Hardware abstraction for Adafruit sensors
ESP8266WiFi Library	Built-in	WiFi connectivity and network management
BlynkSimpleEsp8266	1.3.2	Blynk cloud communication over MQTT

6. System Design & Architecture

6.1 IoT Architecture Overview

The system follows a three-layer IoT architecture:

- Perception Layer: DHT22 sensor captures environmental data (temperature and humidity). RGB LED provides physical status output.
- Processing Layer: NodeMCU ESP8266 firmware reads sensor data, executes automation logic, controls the relay and RGB LED, updates the OLED display, and communicates with the Blynk cloud.
- Application Layer: Blynk IoT cloud receives sensor data and forwards app commands. The Blynk mobile and web dashboard displays live data and sends control signals.

6.2 Complete Pin Mapping

NodeMCU Pin	GPIO #	Connected Component	Function / Notes
D1	GPIO5	OLED SCL	I2C Clock — yellow wire
D2	GPIO4	OLED SDA	I2C Data — green wire
D4	GPIO2	DHT22 DATA	Single-wire temp/humidity protocol
D5	GPIO14	Relay IN	Active LOW — LOW=ON, HIGH=OFF
D6	GPIO12	RGB LED Red	Red channel — appliance OFF indicator
D7	GPIO13	RGB LED Green	Green channel — appliance ON indicator
D8	GPIO15	RGB LED Blue	Blue channel — auto mode indicator
3V3	3.3V	DHT22 VCC + OLED VCC	3.3V regulated power supply
Vin	~5V	Relay VCC	5V from USB — required for relay coil
GND	Ground	All component GNDs	Common ground rail on breadboard

6.3 Blynk IoT Virtual Pin Configuration

Virtual Pin	Data Type	Dashboard Widget	Function
V0	Integer (0/1)	Button — Switch mode	Relay ON/OFF remote control
V1	Double (0–50)	Gauge + Value Display	Live temperature in °C
V2	Double (0–100)	Gauge + Value Display	Live humidity in %
V3	Integer (0/1)	Switch widget	Auto mode ON/OFF toggle
V4	Double (25–40)	Slider widget	Temperature threshold setting

6.4 RGB LED Color Code System

LED Color	System State	Description
RED	Relay OFF (Manual)	Appliance is OFF — system in manual mode, waiting for command
GREEN	Relay ON (Manual)	Appliance turned ON manually via Blynk app button
BLUE	Auto Mode — Standby	Temperature monitoring active, appliance currently OFF
YELLOW (R+G)	Auto Mode — Triggered	Temperature exceeded threshold — appliance auto-activated

7. Working Methodology

Phase 1 — Hardware Assembly

All components were individually tested before integration using dedicated test sketches. The NodeMCU, DHT22, OLED display, relay module, and RGB LED module were then mounted on a cardboard demonstration board and labeled clearly for easy identification during evaluation.

Phase 2 — Sensor Data Acquisition

The DHT22 sensor connected to GPIO2 (D4) is polled every 3 seconds using the Adafruit DHT library. The DHT22 was chosen over the DHT11 for its superior temperature accuracy ($\pm 0.5^{\circ}\text{C}$ vs $\pm 2^{\circ}\text{C}$) and extended measurement range (-40 to $+80^{\circ}\text{C}$). Readings are validated using NaN checks before processing.

Phase 3 — IoT Cloud Integration

The NodeMCU establishes a WiFi connection to a 2.4GHz network and maintains a persistent MQTT connection to the Blynk IoT cloud server. Temperature (V1) and humidity (V2) readings are pushed every 3 seconds to the cloud, appearing live on both the web dashboard and mobile application. The system supports bidirectional communication — the app can send commands which are received via Blynk virtual pin callbacks.

Phase 4 — Appliance Control Logic

The relay module (active LOW) on GPIO14 (D5) controls the connected appliance. Two operating modes are implemented:

- **Manual Mode:** User taps the ON/OFF button widget (V0) in the Blynk app. The relay toggles instantly. RGB LED turns GREEN when ON, RED when OFF. Manual control is blocked when auto mode is active to prevent conflicts.
- **Auto Mode:** Enabled via V3 switch in the app. The system continuously compares current temperature against the threshold set by the V4 slider. When temperature $\geq$ threshold, relay activates automatically (LED turns YELLOW). When temperature drops 2°C below threshold (hysteresis band), relay deactivates. The 2°C hysteresis prevents rapid ON/OFF switching near the threshold.

Phase 5 — Display & Feedback

- **OLED Display:** Renders temperature, humidity, relay state (ON/OFF), and current mode (MANUAL/AUTO) using the U8g2 library. Updates every 2 seconds.
- **RGB LED Module:** Provides color-coded status — no additional resistors needed as the module has built-in current limiting.

8. Code Structure & Explanation

The firmware is written in Arduino C++ and organized into modular, single-responsibility functions for maintainability and readability.

Function / Block	Description
setup()	Initializes all GPIO pins, starts DHT22 and OLED, plays startup animation, connects to WiFi and Blynk cloud, registers BlynkTimer tasks.
loop()	Minimal loop — calls Blynk.run() to maintain cloud connection and timer.run() to execute scheduled sensor readings.
readSensor()	Reads DHT22 every 3 seconds. Validates data. Sends temperature to V1 and humidity to V2. Executes auto-mode logic. Refreshes OLED display.
setRelay(state, source)	Controls relay GPIO (active LOW). Updates RGB LED color based on state + mode. Syncs Blynk V0 button to match physical state.
setRGB(r, g, b)	Sets individual RGB LED channels HIGH or LOW. Used throughout for color-coded status feedback.
updateOLED()	Renders formatted sensor data, relay state, and mode text on SH1106 OLED using U8g2 library with clearBuffer/sendBuffer pattern.
BLYNK_WRITE(V0)	Blynk callback: handles relay toggle from app button. Blocked when auto mode is ON to prevent conflicts with automation logic.
BLYNK_WRITE(V3)	Blynk callback: enables or disables auto mode. Changes LED to BLUE when auto mode activates.
BLYNK_WRITE(V4)	Blynk callback: updates temperature threshold variable when user moves the slider in the app dashboard.

9. Results & Testing

9.1 Individual Component Test Results

#	Component	Test Method	Observed Result	Status
1	NodeMCU ESP8266	Upload blink sketch, check LED	LED blinked correctly	✓ PASS
2	DHT22 Sensor	Run DHT test code, Serial Monitor	T:32.9°C H:58.3% displayed	✓ PASS
3	OLED Display	Upload OLED test code	Text displayed on screen	✓ PASS
4	Relay Module	Upload relay toggle code	Click sound every second	✓ PASS
5	RGB LED Module	Cycle R/G/B in code	Each color displayed correctly	✓ PASS

9.2 Integrated System Test Results

#	Test Case	Expected Result	Actual Result	Status
1	System power ON	Boot screen on OLED, RGB LED color cycle	Startup animation played	✓ PASS
2	DHT22 live readings	Temp + humidity every 3 seconds	32.9°C, 58.3% RH	✓ PASS
3	OLED live display	All values update every 2 seconds	Accurate display confirmed	✓ PASS
4	App relay toggle	Relay ON/OFF, LED green/red	Instant app response	✓ PASS
5	Auto mode activation	LED blue, monitoring starts	Auto mode confirmed	✓ PASS
6	Auto temp trigger	Relay ON when temp > threshold	Yellow LED, relay clicked	✓ PASS
7	Hysteresis check	Relay OFF 2°C below threshold	No chattering observed	✓ PASS
8	Manual blocked in auto	App button ignored	Correctly blocked	✓ PASS
9	WiFi + Blynk connection	Device shows online in app	Connected successfully	✓ PASS

9.3 Live Sensor Readings Recorded

Sample readings captured during testing — DHT22 accuracy verified against a reference thermometer.

Reading #	Temperature (°C)	Humidity (%)	Relay State	Mode
1	32.9	58.3	OFF	Manual
2	32.9	58.3	OFF	Manual
3	32.9	58.4	OFF	Manual
4	33.1	57.9	ON	Auto
5	33.0	58.1	ON	Auto

10. Learning Outcomes

Through this project, the following skills and knowledge areas were developed:

10.1 Embedded Systems & Hardware

- GPIO programming — digital output, input modes, and active-LOW relay control logic
- I2C communication protocol — interfacing SH1106 OLED with only 2 signal wires (SDA + SCL)
- Single-wire DHT22 protocol — temperature and humidity data acquisition
- Component testing methodology — systematic individual testing before integration
- Breadboard circuit assembly and professional hardware presentation

10.2 IoT Architecture & Cloud

- ESP8266 WiFi stack — network connection, SSID/password configuration, 2.4GHz constraints
- MQTT protocol — lightweight publish-subscribe messaging for IoT data transmission
- Blynk IoT platform — template creation, datastream configuration, dashboard design
- Bidirectional cloud communication — device-to-cloud (sensor data) and cloud-to-device (commands)
- Virtual pin architecture — mapping hardware functions to app controls

10.3 Software Engineering

- Modular firmware design — separate functions for each component and responsibility
- State machine logic — tracking relay state, auto mode, and mode transitions
- Hysteresis implementation — preventing relay oscillation near threshold boundaries
- Callback-based event handling — BLYNK_WRITE macros for responsive app control
- Non-blocking architecture — BlynkTimer for scheduled tasks without delay() blocking

11. Scope for Enhancement

The following enhancements are planned for future iterations of this system:

1. Multi-channel expansion: Upgrade to a 4-channel relay to independently control fan, light, AC, and pump from a single NodeMCU.
2. Voice control integration: Connect to Google Assistant or Amazon Alexa via IFTTT webhooks for hands-free appliance control.
3. Energy monitoring: Add ACS712 current sensor to track real-time power consumption per appliance.
4. Light-dependent automation: Integrate an LDR sensor for automatic light control based on ambient brightness levels.
5. Push notifications: Send Blynk app alerts when temperature crosses critical thresholds (e.g. above 40°C).
6. OTA firmware updates: Implement Over-The-Air updates so the system can be upgraded without physical USB access.
7. Local fallback control: Add a local web server so the system remains controllable even when internet connectivity is lost.
8. Data logging: Log sensor readings to Google Sheets via IFTTT for long-term trend analysis and energy optimization

12. Conclusion

All project objectives were successfully achieved. The system was physically built, individually tested, integrated, and demonstrated on a labeled presentation board.

This project demonstrates a complete, end-to-end IoT home automation system built from scratch using the NodeMCU ESP8266 microcontroller. The system successfully achieves remote appliance control via the Blynk mobile application, real-time environmental monitoring with the highly accurate DHT22 sensor ($\pm 0.5^{\circ}\text{C}$), intelligent temperature-based automation with 2°C hysteresis control, live OLED display feedback updated every 2 seconds, and intuitive color-coded RGB LED status indication.

A key aspect of this project was the systematic development methodology: each component was individually verified using dedicated test sketches before integration into the final system. This approach significantly reduced debugging time and ensured reliable operation of the complete system.

The internship at Vaidsys Technologies provided invaluable practical exposure to the complete IoT development lifecycle — spanning embedded firmware development, hardware circuit assembly, cloud platform configuration, and mobile application setup. The skills developed form a strong foundation for advanced IoT, embedded systems, and product development work.

13. References

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